# BRAUER GROUPS OF GENERALISED MARKOFF SURFACES

DASHENG WEI

Abstract. Let $k$ be a field of characteristic zero, let $a, b, c \in k^\times$, and let $m \notin \{0, 4abc\}$. We compute the Brauer group of the smooth cubic surface $(ax^2+by^2+cz^2)w-xyz = mw^3$ and the algebraic Brauer group of its affine complement. The latter is determined by four square classes and may contain elements of order four. A boundary-period calculation gives an explicit finite Galois module computing every torsion subgroup of the complete affine Brauer group, including its transcendental part. The calculation recovers the ordinary Markoff formulas of Colliot-Thélène–Wei–Xu and gives additional descent conditions for coefficient twists. Over $\mathbf{Q}$, the affine Brauer group is algebraic whenever $a, b, c$ do not all have the same square class.

## 1. Introduction

Let $k$ be a field of characteristic zero. Fix $a, b, c \in k^\times$ and $m \in k \setminus \{0, 4abc\}$, and put

$$\Delta = m - 4abc. \tag{1.1}$$

We consider the projective generalised Markoff surface

$$X_m : \quad (ax^2 + by^2 + cz^2)w - xyz = mw^3 \subset \mathbf{P}^3_k \tag{1.2}$$

and its affine part

$$U_m : \quad ax^2 + by^2 + cz^2 - xyz = m. \tag{1.3}$$

The boundary is the split triangle

$$\mathscr{D} = D_x + D_y + D_z, \qquad D_x = \{x = w = 0\}, \quad D_y = \{y = w = 0\}, \quad D_z = \{z = w = 0\}. \tag{1.4}$$

The notation suppresses the fixed coefficients $a, b, c$.

For $a = b = c = 1$, the Brauer groups and the associated integral Brauer–Manin obstruction were studied by Colliot-Thélène, Wei and Xu [1] and by Loughran and Mitankin [2]. Here we retain arbitrary coefficients and distinguish the projective, algebraic affine and complete affine calculations. The coefficient twists change the Galois action on the Picard lattice; in particular, the algebraic affine quotient need not be killed by two.

The projective answer, proved in Theorem 3.1, is especially simple. It is zero modulo constants if one of $a\Delta, b\Delta, c\Delta, m\Delta$ is a square. If all four are nonsquares, it is cyclic of order two, except when their square classes are equal, in which case it is $(\mathbf{Z}/2)^2$. We give quaternion representatives using the conic pencil through $D_x$.

2020 *Mathematics Subject Classification.* Primary 14F22; Secondary 11G35, 14J26.

*Key words and phrases.* Generalised Markoff surfaces, Brauer groups, conic bundles, Galois cohomology.

For the affine algebraic group, let

$$\chi_1 = [a\Delta], \qquad \chi_2 = [b\Delta], \qquad \chi_3 = [c\Delta], \qquad \chi_4 = [m\Delta] \quad \text{in } k^\times/k^{\times 2}. \tag{1.5}$$

Write $N$ for the number of nonzero $\chi_i$ and $d$ for the dimension of their span over $\mathbf{F}_2$. We identify the geometric Picard group with the $D_4$ root lattice, with diagonal action through these four characters. Theorem 4.3 gives

$$\operatorname{Br}_1(U_m)/\operatorname{Br}(k) \simeq \begin{cases} (\mathbf{Z}/2)^{N-d}, & N < 4, \\ (\mathbf{Z}/2)^{5-d}, & N = 4 \text{ and every relation has even weight}, \\ \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{3-d}, & N = 4 \text{ and an odd-weight relation exists}. \end{cases} \tag{1.6}$$

The last case is governed by a relation among three of the four characters. Its order-four generator doubles to a nonzero projective class.

The complete affine Brauer group requires more than square classes: geometrically $\operatorname{Br}(\overline{U}_m) = \mathbf{Q}/\mathbf{Z}(-1)$. In Section 5 we compute the restriction period of the lattice orthogonal to the boundary and construct a free rank-five $\mathbf{Z}/n$-module $T_n$. Theorem 5.2 gives

$$(\operatorname{Br}(U_m)/\operatorname{Br}(k))[n] \simeq \frac{\operatorname{Hom}_{G_k}(T_n, \mathbf{Z}/n)}{\operatorname{Pic}(\overline{U}_m)^{G_k}/n\operatorname{Pic}(\overline{U}_m)^{G_k}}. \tag{1.7}$$

The defining equations, the field of definition and the five-dimensional Galois matrices of $T_n$ are all explicit. Thus (1.7) determines the whole group, including the extension of its algebraic part by its geometric image. We also give an odd-order criterion, recover the ordinary Markoff formula, and prove the assertion over $\mathbf{Q}$ stated in the abstract.

**Notation and conventions.** Fix an algebraic closure $\overline{k}$ of $k$ and put $G_k = \operatorname{Gal}(\overline{k}/k)$. A bar denotes base change to $\overline{k}$. All cohomology of varieties is étale cohomology, and cohomology of fields or Galois modules is continuous Galois cohomology. We use the cohomological Brauer group $\operatorname{Br}(V) = H^2(V, \mathbf{G}_{\mathrm{m}})_{\mathrm{tors}}$ and write

$$\operatorname{Br}_1(V) = \ker[\operatorname{Br}(V) \longrightarrow \operatorname{Br}(\overline{V})].$$

The notation $(f, g)$ denotes a quaternion class. Square classes are written additively, with $[q] = 0$ meaning $q \in k^{\times 2}$. For a torsion group $A$, $A[n]$ denotes its subgroup killed by $n$. The Tate twist $\mathbf{Z}/n(-1)$ means $\operatorname{Hom}(\mu_n, \mathbf{Z}/n)$. The constant Brauer groups inject in the cases considered here, so we identify $\operatorname{Br}(k)$ with its image. Splittings obtained by evaluation at a rational point are not canonical.

## 2. Geometry and descent

**Lemma 2.1.** *The surface $X_m$ is smooth, geometrically rational and has a $k$-point. Moreover, $U_m(k) \neq \varnothing$. Consequently both $\operatorname{Br}(k) \to \operatorname{Br}(X_m)$ and $\operatorname{Br}(k) \to \operatorname{Br}(U_m)$ are split injections, and $\operatorname{Br}(\overline{X}_m) = 0$.*

*Proof.* On the boundary the derivatives with respect to $x, y, z$ are $-yz, -xz, -xy$. Their simultaneous vanishing leaves only the three vertices of the triangle, at which the derivative

with respect to $w$ is respectively $a, b, c$. Thus there are no singular points at infinity. In the affine chart a singular point satisfies

$$2ax = yz, \qquad 2by = xz, \qquad 2cz = xy.$$

If a coordinate vanishes, all three vanish, forcing $m = 0$. Otherwise these equations give $x^2 = 4bc$, $y^2 = 4ac$, $z^2 = 4ab$, and $xyz = 8abc$, so the surface equation forces $m = 4abc$. Both possibilities are excluded.

Every smooth cubic surface over an algebraically closed field is rational; see, for example, [4]. The boundary vertex $[1 : 0 : 0 : 0]$ is a $k$-point. One can also construct an affine point directly. Choose $q \in k^\times$ with $bq^2 \neq c$, and set

$$t = bq + c/q, \qquad R = m - at^2, \qquad z = \frac{R - b}{bq - c/q}, \qquad y = 1 + qz, \qquad x = t.$$

Then

$$by^2 + cz^2 - tyz = (y - qz)(by - cz/q) = R,$$

so $(x, y, z) \in U_m(k)$. Evaluation at these points splits the constant Brauer groups. Finally, geometric rationality gives $\mathrm{Br}(\overline{X}_m) = 0$. □

**Lemma 2.2.** *The geometric units of $U_m$ are constant and $\mathrm{Pic}(\overline{U}_m)$ is a free abelian group of rank four. For $V = X_m$ or $U_m$ one has*

$$\mathrm{Pic}(V) = \mathrm{Pic}(\overline{V})^{G_k}, \qquad \mathrm{Br}_1(V)/\mathrm{Br}(k) = H^1(k, \mathrm{Pic}(\overline{V})).$$

*For every $n \geq 1$,*

$$H^2(U_m, \mu_n)/H^2(k, \mu_n) \simeq H^2(\overline{U}_m, \mu_n)^{G_k}. \tag{2.1}$$

*Proof.* The divisor sequence for $\overline{U}_m \subset \overline{X}_m$ is

$$0 \longrightarrow \overline{k}[U_m]^\times/\overline{k}^\times \longrightarrow \bigoplus_{i=x,y,z} \mathbf{Z}[D_i] \longrightarrow \mathrm{Pic}(\overline{X}_m) \longrightarrow \mathrm{Pic}(\overline{U}_m) \longrightarrow 0.$$

The boundary intersection matrix has diagonal entries $-1$ and off-diagonal entries 1; its determinant is 4. The boundary classes are therefore independent and the units are constant. The explicit blow-up basis in Section 4 shows that their quotient is free of rank four.

For the Picard and algebraic Brauer assertions, apply Hochschild–Serre to $\mathbf{G}_\mathrm{m}$. A rational point makes the pullback $H^j(k, \mathbf{G}_\mathrm{m}) \to H^j(V, \mathbf{G}_\mathrm{m})$ injective in every degree, so the usual obstruction maps vanish. This gives the stated identifications. For (2.1), Kummer gives $H^1(\overline{U}_m, \mu_n) = 0$ because the units are constant and the geometric Picard group has no torsion. The sole possible differential out of $H^2(\overline{U}_m, \mu_n)^{G_k}$ is then the $d_3$ into $H^3(k, \mu_n)$; it vanishes because the rational point also splits this pullback. The edge sequence is (2.1). We use these standard descent and Kummer constructions in the form explained in [3]. □

## 3. The Brauer group of the projective surface

Throughout this section put $\Delta = m - 4abc$ and let $t = x/w$. The distinction between the projective surface and its affine open subset is essential: the following calculation concerns classes unramified along the three lines at infinity as well as along every affine divisor.

**Theorem 3.1.** *The Brauer group of $X_m$, modulo constant classes, is*

$$\mathrm{Br}(X_m)/\mathrm{Br}(k) \cong \begin{cases} 0, & \text{if at least one of } a\Delta, b\Delta, c\Delta, m\Delta \text{ belongs to } k^{*2}, \\ (\mathbf{Z}/2)^2, & \text{if } ab, ac, am \in k^{*2} \text{ and } a\Delta \notin k^{*2}, \\ \mathbf{Z}/2, & \text{otherwise.} \end{cases}$$

*In every nontrivial case the quaternion class*

$$B = (t^2 - 4bc, b\Delta)$$

*belongs to* $\mathrm{Br}(X_m)$ *and is nonconstant. It generates the quotient when that quotient is cyclic. In the second case choose $u, v \in k^*$ with $u^2 = bc$ and $v^2 = m/a$. Then the classes*

$$B = (t^2 - 4bc, b\Delta), \qquad C = ((t - 2u)(t - v), b\Delta)$$

*generate* $\mathrm{Br}(X_m)/\mathrm{Br}(k)$*. Both classes evaluate to zero at $P = [1 : 0 : 0 : 0]$. Consequently evaluation at $P$ gives a splitting of the constant subgroup, and* $\mathrm{Br}(X_m)$ *is the direct sum of* $\mathrm{Br}(k)$ *and the finite group displayed above.*

We first describe the conic pencil and its splitting characters. Let $L_x$ be the line $x = w = 0$ on $X_m$. The planes $x = tw$ through $L_x$, after removal of their fixed component $L_x$, define the conic-bundle morphism

$$f : X_m \longrightarrow \mathbf{P}^1_k.$$

On $w \neq 0$ this morphism is given by the function $t = x/w$. Its finite fibres are the residual conics

$$by^2 + cz^2 - tyz + (at^2 - m)w^2 = 0. \tag{3.1}$$

The fibre at infinity is the union of the two $k$-lines $y = w = 0$ and $z = w = 0$. In particular this fibre is split. The two quadratics

$$F_1(t) = t^2 - 4bc, \qquad F_2(t) = t^2 - m/a$$

are separable and have disjoint zero sets. Their four geometric zeros give all finite degenerate fibres of the pencil.

**Lemma 3.2.** *Let $Q$ be a closed point of the finite base. If $F_1(Q) = 0$, the two geometric components of $f^{-1}(Q)$ are defined over $k(Q)(\sqrt{b\Delta})$. If $F_2(Q) = 0$, they are defined over $k(Q)(\sqrt{a\Delta})$. The corresponding generic conic has Brauer class*

$$A = (F_1, b(m - at^2)) = (F_1, b\Delta) + (F_2, a\Delta) \quad \text{in } \mathrm{Br}(k(t)). \tag{3.2}$$

*Proof.* Multiplying (3.1) by $4b$ and completing the square gives

$$(2by - tz)^2 = F_1 z^2 + 4b(m - at^2)w^2.$$

At a zero of $F_1$ this becomes $(2by - tz)^2 = 4b\Delta w^2$, proving the first assertion. At a zero of $F_2$ the equation is binary in $y, z$, and its discriminant is $F_1(Q) = \Delta/a$, whose square class is $a\Delta$. This proves the second assertion and the first expression for $A$.

For the final identity observe that $F_1 - F_2 = \Delta/a$. The quaternion identity

$$(r, r - s) = (r, -s) + (s, r - s)$$

follows from the Steinberg relation. Applying it with $r = F_1$ and $s = \Delta/a$ gives $(F_1, F_2) = (F_1, -\Delta/a) + (a\Delta, F_2)$. Since $m - at^2 = -aF_2$, adding $(F_1, -ab)$ proves (3.2). □

Let $\mathcal{N}$ be the set of nonsplit finite degenerate closed fibres. For $Q \in \mathcal{N}$ write

$$d_Q = \begin{cases} b\Delta, & F_1(Q) = 0, \\ a\Delta, & F_2(Q) = 0. \end{cases}$$

Thus $d_Q$ represents a nonzero quadratic character of $k(Q)$. Identify quadratic characters with square classes and set

$$R = \ker \left[ \mathbf{F}_2^{\mathcal{N}} \longrightarrow k^*/k^{*2}, \quad (\epsilon_Q)_Q \longmapsto \prod_{Q \in \mathcal{N}} N_{k(Q)/k}(d_Q)^{\epsilon_Q} \right]. \tag{3.3}$$

**Proposition 3.3.** *If $\mathcal{N} \neq \varnothing$, there is an isomorphism*

$$\mathrm{Br}(X_m)/\mathrm{Br}(k) \cong R/\langle \mathbf{1} \rangle, \qquad \mathbf{1} = (1)_{Q \in \mathcal{N}}.$$

*If $\mathcal{N} = \varnothing$, then $\mathrm{Br}(X_m) = \mathrm{Br}(k)$.*

*Proof.* Write $K = k(t)$ and let $X_\eta$ be the generic conic. The map $\mathrm{Br}(K) \to \mathrm{Br}(X_\eta)$ is surjective, with kernel generated by $A$. Hence any class on $X_m$ is represented on its generic fibre by a class $\beta \in \mathrm{Br}(K)$. Purity and compatibility of residues with pullback describe precisely which such representatives extend to $X_m$.

At a smooth fibre the residue of $\beta$ must be zero: the fibre is geometrically integral, so restriction of characters from its residue field to its function field is injective. The same conclusion holds at a split degenerate fibre, by using either component. At a nonsplit degenerate fibre the residue may be zero or the quadratic character $d_Q$; these are exactly the characters that become trivial over the constant field of its components. The split fibre at infinity requires zero residue there. Horizontal divisors produce no further conditions, because their valuations are trivial on the base field $K$.

The Faddeev exact sequence says that a collection of residues on $\mathbf{P}^1_k$ comes from $\mathrm{Br}(K)$ precisely when its total corestriction is zero, and that the kernel of all residues is $\mathrm{Br}(k)$. It therefore identifies the permitted residue collections with $R$. Exactness here holds over every field of characteristic zero; no cohomological-dimension assumption is needed. In this particular calculation one can also construct the representatives directly. If $p_Q \in k[t]$ is the monic irreducible polynomial of $Q$, the class

$$\beta_\epsilon = \sum_{Q \in \mathcal{N}} \epsilon_Q (p_Q(t), d_Q)$$

has exactly the prescribed finite residues. Its residue at infinity is their total corestriction, and therefore vanishes for $\epsilon \in R$. Its pullback extends to $X_m$ by purity.

The residues of $A$ are precisely $d_Q$ at every member of $\mathcal{N}$, and zero elsewhere. Thus killing the generic conic class identifies residue vectors differing by $\mathbf{1}$. Conversely, if a pullback is constant, subtracting that constant leaves a representative in the kernel generated by $A$; there are no additional relations. If $\mathcal{N}$ is empty, every permitted representative is already constant. This proves both assertions. □

*Proof of Theorem 3.1.* The fibres over the zeros of $F_1$ are split exactly when $b\Delta$ or $c\Delta$ is a square in $k$. Indeed, if $bc$ is a square, both zeros are rational and the criterion is simply $b\Delta \in k^{*2}$. Otherwise their residue field is $k(\sqrt{bc})$, and a scalar from $k^*$ becomes a square in this quadratic field exactly when its square class is 1 or $bc$. Similarly the fibres over the zeros of $F_2$ are split exactly when $a\Delta$ or $m\Delta$ is a square.

If either pair is split, all nonsplit fibres belong to the other pair. That pair contributes either one quadratic closed point or two rational points with the same character. In both cases its permitted residue kernel is generated by its full residue vector, which disappears in Proposition 3.3. This proves the first case of the theorem.

Suppose now that neither pair is split. An irreducible quadratic contributes one coordinate to $R$ and no corestriction condition, since $N_{k(Q)/k}(d_Q) = d_Q^2$. A rational pair contributes two coordinates and one condition given by their common nonzero character. If both quadratics are irreducible, $R$ has dimension two; if exactly one splits into rational points, it has dimension two again. After quotienting by $\mathbf{1}$, the answer in each case is $\mathbf{Z}/2$.

If both pairs are rational, the corestriction map has rank one when $[b\Delta] = [a\Delta]$ and rank two otherwise. Its kernel therefore has dimension three or two, respectively. Equality of these two nonzero characters means $ab \in k^{*2}$. Together with $bc, m/a \in k^{*2}$, this is equivalent to $ab, ac, am \in k^{*2}$, yielding the asserted rank-two case.

Finally, $B$ has the full residue vector on the first pair and zero vector on the second. Whenever neither pair is split this vector survives the quotient by $\mathbf{1}$. In the rank-two case $C$ has residue $b\Delta$ at $t = 2u$ and $t = v$, and no other residue; its vector is independent of that of $B$ modulo $\mathbf{1}$. These descriptions also prove that both classes extend. Near infinity their first entries are, modulo squares, respectively $1 - 4bc/t^2$ and $(1 - 2u/t)(1 - v/t)$. Both specialize to 1. Thus their evaluations at $P$, which lies on the fibre at infinity, are zero, as claimed. □

**Remark 3.4.** Although the calculation uses the $x$-pencil, its final answer is symmetric in $a, b, c$. The rank-two condition says that $a, b, c, m$ have the same square class. If this condition holds, one may choose the square roots of $bc, ac, ab$ compatibly and make the change of variables

$$X = x/\sqrt{bc}, \qquad Y = y/\sqrt{ac}, \qquad Z = z/\sqrt{ab}.$$

The affine equation becomes $X^2 + Y^2 + Z^2 - XYZ = m/(abc)$, so this case also agrees directly with the usual Markoff calculation. Such a change of variables need not be defined over $k$ in the remaining cases; the residue argument is what permits arbitrary coefficients without adjoining these square roots.

More generally, Proposition 3.3 gives representatives for every projective Brauer class, rather than only its abstract group. One first chooses a basis of the elementary residue kernel modulo its full vector and then uses the corresponding sums $\beta_\epsilon$. All ingredients are square classes in $k$ or in the two indicated quadratic residue fields. In particular, the calculation makes no assumption that $k$ is a number field, that its constant Brauer group vanishes, or that the generic conic has a rational point. The only generic-conic relation to remove is its own class $A$.

## 4. The Picard lattice and the algebraic Brauer group

Throughout this section, put $\Delta = m - 4abc$ and assume $abcm\Delta \neq 0$. For $q \in k^\times$, write $[q]$ for its class in $k^\times/k^{\times 2}$. We identify this square-class group with $H^1(k, \mathbf{Z}/2)$, and write its operation additively.

4.1. **The geometric configuration.** Choose $r_1, r_2 \in \overline{k}$ with $r_1^2 = bc$ and $r_2^2 = ac$, and put $r_3 = r_1 r_2/c$. Thus $r_3^2 = ab$ and $r_1 r_2 r_3 = abc$. Set

$$X = x/r_1, \qquad Y = y/r_2, \qquad Z = z/r_3, \qquad M = m/(abc), \qquad D = \Delta/(abc).$$

In these coordinates the equation is

$$(X^2 + Y^2 + Z^2)w - XYZ = Mw^3, \qquad D = M - 4. \tag{4.1}$$

Fix $u, v \in \overline{k}$ with $u^2 = M$ and $v^2 = D$. Besides the three boundary lines $L_1 = \{X = w = 0\}$, $L_2 = \{Y = w = 0\}$ and $L_3 = \{Z = w = 0\}$, the surface has the following twenty-four lines, where $\epsilon, \delta \in \{1, -1\}$:

$$\begin{aligned}
l_1(\epsilon,\delta):&\quad X = 2\epsilon w, && Y - \epsilon Z = \delta v w,\\
l_2(\epsilon,\delta):&\quad Y = 2\epsilon w, && Z - \epsilon X = \delta v w,\\
l_3(\epsilon,\delta):&\quad Z = 2\epsilon w, && X - \epsilon Y = \delta v w,\\
l_4(\epsilon,\delta):&\quad X = \epsilon u w, && Y = (\epsilon u + \delta v)Z/2,\\
l_5(\epsilon,\delta):&\quad Y = \epsilon u w, && Z = (\epsilon u + \delta v)X/2,\\
l_6(\epsilon,\delta):&\quad Z = \epsilon u w, && X = (\epsilon u + \delta v)Y/2.
\end{aligned}$$

These equations describe the generalized lines explicitly after the indicated linear substitution. Substitution into (4.1) verifies that they lie on the surface. Their incidence configuration is the standard one recorded in [1, Lemma 3.1]. In particular, the six lines $e_i = l_i(1,1)$ are mutually disjoint. Blowing them down over $\overline{k}$ gives $\mathbf{P}^2$, and hence

$$\mathrm{Pic}(\overline{X}_m) = \mathbf{Z}h \oplus \bigoplus_{i=1}^{6} \mathbf{Z}e_i, \qquad L_j = h - e_j - e_{j+3} \quad (1 \le j \le 3),$$

where $h$ is the pullback of a line in $\mathbf{P}^2$.

The three classes $L_j$ are linearly independent: intersecting a relation $\sum_j n_j L_j = 0$ with the three boundary lines gives a system with diagonal entries $-1$ and off-diagonal entries 1, whose determinant is 4. The divisor exact sequence for $U_m = X_m \setminus (L_1 \cup L_2 \cup L_3)$ therefore gives

$$\overline{k}[U_m]^\times = \overline{k}^\times, \qquad \mathrm{Pic}(\overline{U}_m) = \mathrm{Pic}(\overline{X}_m)/\langle L_1, L_2, L_3\rangle. \tag{4.2}$$

The latter quotient is free with basis $e_1, e_2, e_3, e_4$, since its defining relations eliminate $h, e_5, e_6$ as follows:

$$h = e_1 + e_4, \qquad e_5 = e_1 + e_4 - e_2, \qquad e_6 = e_1 + e_4 - e_3. \tag{4.3}$$

**Proposition 4.1.** *There is an isomorphism of Galois lattices*

$$\mathrm{Pic}(\overline{U}_m) \simeq \mathcal{L} := \{(z_1, z_2, z_3, z_4) \in \mathbf{Z}^4 : z_1 + z_2 + z_3 + z_4 \equiv 0 \pmod 2\},$$

*under which Galois acts diagonally through the quadratic characters*

$$\chi_1 = [a\Delta], \quad \chi_2 = [b\Delta], \quad \chi_3 = [c\Delta], \quad \chi_4 = [m\Delta]. \tag{4.4}$$

*Thus its splitting field is*

$$K = k(\sqrt{a\Delta}, \sqrt{b\Delta}, \sqrt{c\Delta}, \sqrt{m\Delta}) = k(\sqrt{ab}, \sqrt{ac}, \sqrt{am}, \sqrt{a\Delta}).$$

*Proof.* We give the action explicitly, including the coefficient twists. Let $\sigma$ change the sign of $v$, let $\tau$ change the sign of $u$, and let $p$ and $q$ denote the coordinate transformations $(X, Y, Z) \mapsto (-X, Y, -Z)$ and $(X, Y, Z) \mapsto (X, -Y, -Z)$, respectively. Each Galois element acts as a product of these transformations, according to its signs on $r_1, r_2, u, v$. The four formal transformations commute; a particular field may impose relations among them. On the basis of (4.3), their actions are

| | $e_1$ | $e_2$ | $e_3$ | $e_4$ |
|---|---|---|---|---|
| $\sigma$ | $-e_1$ | $-e_2$ | $-e_3$ | $-e_4$ |
| $\tau$ | $e_1$ | $e_2$ | $e_3$ | $-e_1 + e_2 + e_3 - e_4$ |
| $p$ | $e_3 - e_2$ | $-e_2$ | $e_1 - e_2$ | $e_1 - e_2 - e_3 + e_4$ |
| $q$ | $-e_1$ | $e_3 - e_1$ | $e_2 - e_1$ | $e_4$ |

For example, the equations of the lines give $p(e_1) = l_1(-1, 1)$ and $p(e_4) = l_4(-1, -1)$. In the blow-up basis these classes are $h - e_2 - e_6$ and $h - e_2 - e_3$, respectively, which give the displayed entries after (4.3). The other entries follow from the same line equations and the usual classes of the twenty-seven lines.

The following four vectors diagonalize this action over $\mathbf{Q}$:

$$W_1 = -e_1 + e_2 + e_3, \qquad W_2 = e_1 - e_2 + e_3,$$
$$W_3 = e_1 + e_2 - e_3, \qquad W_4 = e_1 - e_2 - e_3 + 2e_4.$$

Their sign characters, in the same order, are $p\sigma, q\sigma, pq\sigma, \tau\sigma$. Since the signs of $r_1, r_2, u, v$ correspond to the square classes $[bc], [ac], [M], [D]$, these characters are exactly (4.4). Indeed, $[bcD] = [a\Delta]$, $[acD] = [b\Delta]$, $[bc \cdot ac \cdot D] = [c\Delta]$ and $[MD] = [m\Delta]$.

Put $E_i = W_i/2$. Then

$$e_1 = E_2 + E_3, \quad e_2 = E_1 + E_3, \quad e_3 = E_1 + E_2, \quad e_4 = E_1 + E_4.$$

The four coefficient vectors on the right have determinant of absolute value 2. They belong to the even-sum sublattice of $\mathbf{Z}^4$, which also has index 2, so they generate precisely that sublattice. Finally, multiplying the first square class in (4.4) by each of the other three gives $[ab], [ac], [am]$, proving the equality of the two displayed fields. □

**Lemma 4.2.** *Let $\Lambda \subset \mathrm{Pic}(\overline{X}_m)$ be the subgroup orthogonal to the three boundary lines. Intersection identifies $\Lambda$ with*

$$\mathcal{L}^\vee = \mathbf{Z}^4 + \mathbf{Z}\omega, \qquad \omega = (1, 1, 1, 1)/2.$$

*If $\varepsilon_i$ are the four standard coordinate vectors of this dual lattice, their corresponding divisor classes are*

$$\varepsilon_1 = h - e_2 - e_3 - e_4, \qquad \varepsilon_2 = h - e_1 - e_3 - e_5,$$
$$\varepsilon_3 = h - e_1 - e_2 - e_6, \qquad \varepsilon_4 = h - e_4 - e_5 - e_6,$$
$$\omega = 2h - e_1 - e_2 - e_3 - e_4 - e_5 - e_6.$$

*In particular, $\Lambda$ has the basis*

$$A_i = \omega - \varepsilon_i - \varepsilon_4 = e_{i+3} - e_i \quad (1 \le i \le 3), \qquad A_4 = -\varepsilon_4 = e_4 + e_5 + e_6 - h.$$

*Proof.* The intersection lattice of the blow-up of $\mathbf{P}^2$ is unimodular, and (4.2) is a torsion-free quotient. Its integral dual is consequently the orthogonal complement of the boundary classes. The dual of the even-sum lattice is the displayed lattice $\mathcal{L}^\vee$. The formulas follow by pairing the indicated classes with $e_1, e_2, e_3, e_4$, which represent $E_2 + E_3, E_1 + E_3, E_1 + E_2, E_1 + E_4$, respectively. For completeness, a class $u_0 h + \sum_{i=1}^{6} v_i e_i$ lies in $\Lambda$ precisely when $v_{j+3} = -u_0 - v_j$ for $1 \le j \le 3$; these relations verify the formulas directly. The four vectors $A_i$ have determinant $-1$ in the basis $\varepsilon_1, \varepsilon_2, \varepsilon_3, \omega$, so they form a basis. □

4.2. **Computation of the algebraic group.** Let $N$ denote the number of nonzero characters in (4.4), and put

$$d = \dim_{\mathbf{F}_2} \langle \chi_1, \chi_2, \chi_3, \chi_4 \rangle.$$

A relation has *odd weight* if it is of the form $\sum_i \epsilon_i \chi_i = 0$ with $\sum_i \epsilon_i = 1$ in $\mathbf{F}_2$.

**Theorem 4.3.** *The algebraic Brauer group is given by*

$$\operatorname{Br}_1(U_m)/\operatorname{Br}(k) \simeq \begin{cases} (\mathbf{Z}/2)^{N-d}, & N < 4, \\ (\mathbf{Z}/2)^{5-d}, & N = 4 \text{ and no odd-weight relation exists}, \\ \mathbf{Z}/4 \oplus (\mathbf{Z}/2)^{3-d}, & N = 4 \text{ and an odd-weight relation exists}. \end{cases}$$

*In the last case $d \in \{2, 3\}$. If $N = 4$, an odd-weight relation exists precisely when at least one of*

$$abc\Delta, \qquad abm\Delta, \qquad acm\Delta, \qquad bcm\Delta$$

*is a square in $k$. In particular, the possible algebraic quotients are $0$, $\mathbf{Z}/2$, $(\mathbf{Z}/2)^2$, $(\mathbf{Z}/2)^3$, $(\mathbf{Z}/2)^4$, $\mathbf{Z}/4$ and $\mathbf{Z}/4 \oplus \mathbf{Z}/2$.*

*Proof.* By Lemmas 2.1 and 2.2, Hochschild–Serre gives

$$\operatorname{Br}_1(U_m)/\operatorname{Br}(k) \simeq H^1(k, \operatorname{Pic}(\overline{U}_m)).$$

This argument applies to every field of characteristic zero.

Write $\mathcal{M} = \bigoplus_{i=1}^{4} \mathbf{Z}(\chi_i)$ for the diagonal sign lattice containing $\mathcal{L}$. Reduction of the sum of the coordinates modulo 2 gives an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \mathcal{M} \longrightarrow \mathbf{Z}/2 \longrightarrow 0, \tag{4.5}$$

where the last module is trivial. A nontrivial sign character contributes one copy of $\mathbf{Z}/2$ to $H^1(k,\mathcal{M})$, while a trivial character contributes zero. Under this identification, the map

$$H^1(k,\mathcal{M}) = (\mathbf{F}_2)^N \longrightarrow H^1(k,\mathbf{Z}/2)$$

sends the $i$th basis vector to $\chi_i$. For instance, a generator for the $i$th sign summand is the cocycle $g \mapsto -\chi_i(g)$, where the values of $\chi_i$ are represented by 0 and 1 as integers.

If $N < 4$, a standard coordinate vector belonging to a trivial sign summand is invariant and maps to 1 in $\mathbf{Z}/2$. The invariant map in (4.5) is therefore surjective, and $H^1(k,\mathcal{L})$ is the kernel of the preceding character map. Its dimension is $N - d$, proving the first case.

Suppose now that $N = 4$. Then $\mathcal{M}^{G_k} = 0$, and the same sequence gives

$$0 \longrightarrow \mathbf{Z}/2\langle\kappa\rangle \longrightarrow H^1(k,\mathcal{L}) \longrightarrow R \longrightarrow 0, \qquad R = \ker\big((\mathbf{F}_2)^4 \xrightarrow{(\chi_i)} H^1(k,\mathbf{Z}/2)\big). \tag{4.6}$$

Here $\dim R = 4 - d$. To determine the extension, take $\epsilon \in R$ and consider the integer-valued cocycle

$$c_\epsilon(g) = -\big(\epsilon_1\chi_1(g), \epsilon_2\chi_2(g), \epsilon_3\chi_3(g), \epsilon_4\chi_4(g)\big).$$

The relation defining $R$ makes its coordinate sum even, so this is a cocycle with values in $\mathcal{L}$. Twice this cocycle is the coboundary $g\epsilon - \epsilon$. If $\epsilon$ has even weight, then $\epsilon \in \mathcal{L}$ and its class has order dividing 2. If $\epsilon$ has odd weight, its image in $\mathcal{M}/\mathcal{L}$ is 1, and consequently

$$2[c_\epsilon] = \kappa. \tag{4.7}$$

Thus, if all relations have even weight, the group in (4.6) is elementary abelian of dimension $5 - d$. Otherwise choose an odd relation as one basis vector of $R$, and make all the other basis vectors even by adding it when necessary. Its lift has order 4 by (4.7); the remaining $3 - d$ lifts have order 2 and are independent. This proves the last case.

Since all four characters are nonzero, an odd relation must involve exactly three of them. Such a relation spans a two-dimensional space, and the fourth character increases its dimension by at most one. Taking the products of the corresponding square classes gives exactly the four square conditions stated in the theorem. □

**Remark 4.4.** The proof specifies generators as Galois cocycles, including the order-four cases. Equal-character pair relations have especially simple quaternion representatives. In the notation of (4.4), they are

$$\begin{array}{r|l} \chi_1 = \chi_2 & (z - 2\sqrt{ab}, a\Delta) \\ \chi_1 = \chi_3 & (y - 2\sqrt{ac}, a\Delta) \\ \chi_2 = \chi_3 & (x - 2\sqrt{bc}, b\Delta) \\ \chi_1 = \chi_4 & (x - \sqrt{m/a}, a\Delta) \\ \chi_2 = \chi_4 & (y - \sqrt{m/b}, b\Delta) \\ \chi_3 = \chi_4 & (z - \sqrt{m/c}, c\Delta). \end{array}$$

Each square root belongs to $k$ under its indicated equality. The finite zero divisor consists of two lines interchanged by the second entry of the symbol; hence the corresponding quaternion is unramified on $U_m$. When $N = 4$, the connecting class $\kappa$ is represented by $(x^2 - 4bc, b\Delta)$, the nontrivial projective class computed above. An odd relation therefore

gives an algebraic class whose double is that projective class. It cannot be represented by a quaternion, because its order modulo constants is four.

## 5. The complete affine Brauer group

Throughout this section, $\Delta = m - 4abc \neq 0$ and $m \neq 0$. We give a finite calculation of every torsion subgroup of the complete Brauer group, including its transcendental part. Keeping the Galois action in this calculation is essential: a calculation after adjoining square roots does not, by itself, settle descent to $k$.

Choose $q_a, q_b, q_c, q_m \in \overline{k}$ with

$$q_a^2 = a\Delta, \qquad q_b^2 = b\Delta, \qquad q_c^2 = c\Delta, \qquad q_m^2 = m\Delta,$$

and put

$$L = k(q_a, q_b, q_c, q_m), \quad M = \frac{m}{abc}, \quad D = \frac{\Delta}{abc}, \quad r = \frac{\Delta q_m}{q_a q_b q_c}, \quad s = \frac{\Delta^2}{q_a q_b q_c}, \quad \eta = \frac{r-s}{2}.$$

Thus $r^2 = M$, $s^2 = D$, $M - D = 4$, and

$$\eta^{-1} = \frac{r+s}{2}. \tag{5.1}$$

Denote the four sign characters by

$$\epsilon_i : G_k \longrightarrow \{1, -1\} \qquad (1 \leq i \leq 4).$$

They correspond to $a\Delta, b\Delta, c\Delta, m\Delta$, respectively.

5.1. **The boundary lattice and its period.** Let $\mathscr{D} = D_x + D_y + D_z = X_m \setminus U_m$ be the boundary triangle. Its three components and its three vertices are defined over $k$. Set

$$\Lambda = \{\lambda \in \mathrm{Pic}(\bar{X}_m) : \lambda \cdot D_x = \lambda \cdot D_y = \lambda \cdot D_z = 0\}.$$

Restriction to the boundary defines a homomorphism

$$\phi : \Lambda \longrightarrow \mathrm{Pic}^0(\bar{\mathscr{D}}).$$

The chosen cyclic ordering of the three components identifies $\mathrm{Pic}^0(\bar{\mathscr{D}})$ with $\overline{k}^*$; reversing the ordering inverts $\phi$.

**Proposition 5.1.** *As a Galois lattice, $\Lambda$ has the following coordinate model:*

$$\Lambda = \mathbb{Z}e_1 + \mathbb{Z}e_2 + \mathbb{Z}e_3 + \mathbb{Z}e_4 + \mathbb{Z}h, \qquad 2h = e_1 + e_2 + e_3 + e_4.$$

*Here the $e_i$ are independent in $\Lambda \otimes \mathbb{Q}$ and $g(e_i) = \epsilon_i(g)e_i$. The period is*

$$\phi(e_1) = \phi(e_2) = \phi(e_3) = -1, \qquad \phi(e_4) = -\eta^2, \qquad \phi(h) = \eta. \tag{5.2}$$

*Moreover,*

$$\mathrm{Pic}(\bar{U}_m) \simeq \Lambda^\vee := \mathrm{Hom}(\Lambda, \mathbb{Z}).$$

*Proof.* The normalization

$$X = \frac{x}{q_b q_c/\Delta}, \qquad Y = \frac{y}{q_a q_c/\Delta}, \qquad Z = \frac{z}{q_a q_b/\Delta}$$

identifies $U_L$ with $X^2 + Y^2 + Z^2 - XYZ = M$. On its projective compactification choose the six lines

$$\begin{aligned} &E_1 : X = 2w, \quad Y - Z = -sw, && E_4 : X = rw, \quad Y = \eta Z, \\ &E_2 : Y = 2w, \quad Z - X = -sw, && E_5 : Y = rw, \quad Z = \eta X, \\ &E_3 : Z = 2w, \quad X - Y = -sw, && E_6 : Z = rw, \quad X = \eta Y. \end{aligned}$$

For this period calculation we use the independent marking by the six lines $l_i(1,-1)$, rather than the marking by $l_i(1,1)$ that can be used in the Picard-lattice calculation. These are six mutually disjoint lines used in [1]. Their simultaneous contraction is $\mathbb{P}^2_L$. If $H$ denotes the pullback of a line in this plane, then $H, E_1, \ldots, E_6$ form a basis of $\mathrm{Pic}(\bar{X}_m)$, with intersection form $\mathrm{diag}(1,-1,\ldots,-1)$, and

$$D_x = H - E_1 - E_4, \quad D_y = H - E_2 - E_5, \quad D_z = H - E_3 - E_6.$$

Consequently a basis of $\Lambda$ is

$$\alpha_i = E_i - E_{i+3} \quad (1 \le i \le 3), \qquad \alpha_4 = H - E_4 - E_5 - E_6. \tag{5.3}$$

We spell out the period computation. On the original boundary components use the cyclic coordinates

$$t_x = y/z, \qquad t_y = z/x, \qquad t_z = x/y.$$

There are nonzero $\kappa_x, \kappa_y, \kappa_z \in L$ with $\kappa_x\kappa_y\kappa_z = 1$ such that the intersection points of $E_i, E_{i+3}$ on the $i$-th boundary component have coordinates $\kappa_i, \kappa_i\eta$. After contraction, a general plane line meets the boundary in three points whose coordinate product is $-\eta$. To determine this product without choosing contraction coordinates, take the line of class $H - E_1 - E_2$. Its inverse image is the normalized Markoff line

$$Z = -rw, \qquad X = -\eta Y.$$

Its three intersection coordinates in the contracted plane are $\kappa_x, \kappa_y, -\kappa_z\eta$, whose product is $-\eta$. The usual description of $\mathrm{Pic}^0$ of a triangle by the product of its gluing parameters now gives

$$\phi(\alpha_i) = \eta^{-1} \quad (1 \le i \le 3), \qquad \phi(\alpha_4) = -\eta^{-2}.$$

Equivalently, for $dH - \sum r_i E_i \in \Lambda$, so that $r_i + r_{i+3} = d$, its period is

$$(-\eta)^d \prod_{i=1}^{3} \kappa_i^{-r_i} (\kappa_i \eta)^{-r_{i+3}}.$$

Define

$$\begin{aligned} &e_1 = \alpha_2 + \alpha_3 - \alpha_4, && e_2 = \alpha_1 + \alpha_3 - \alpha_4, \\ &e_3 = \alpha_1 + \alpha_2 - \alpha_4, && e_4 = -\alpha_4, \qquad h = \alpha_1 + \alpha_2 + \alpha_3 - 2\alpha_4. \end{aligned}$$

These elements generate $\Lambda$ and have the sole relation $2h = \sum e_i$. They give (5.2). For completeness, the following table records the action on (5.3). A row denotes a sign change of the indicated square root, with the other three signs fixed; if the square classes are dependent, restrict these formulas to the sign tuples actually occurring in $\mathrm{Gal}(L/k)$. Put $A = \alpha_1 + \alpha_2 + \alpha_3 - \alpha_4$.

| | $\alpha_1$ | $\alpha_2$ | $\alpha_3$ | $\alpha_4$ |
|---|---|---|---|---|
| $q_a$ | $A$ | $-\alpha_3 + \alpha_4$ | $-\alpha_2 + \alpha_4$ | $\alpha_4$ |
| $q_b$ | $-\alpha_3 + \alpha_4$ | $A$ | $-\alpha_1 + \alpha_4$ | $\alpha_4$ |
| $q_c$ | $-\alpha_2 + \alpha_4$ | $-\alpha_1 + \alpha_4$ | $A$ | $\alpha_4$ |
| $q_m$ | $\alpha_1 - \alpha_4$ | $\alpha_2 - \alpha_4$ | $\alpha_3 - \alpha_4$ | $-\alpha_4$ |

The table follows by applying the sign changes to the displayed line equations and expressing the resulting lines in the $H, E_i$ basis. For example, changing $q_a$ fixes $E_1, E_5, E_6$ and sends $E_4$ to the line $X = -rw$, $Y = -\eta Z$, of class $H - E_2 - E_3$. The table gives the asserted diagonal action on the $e_i$.

Finally the boundary classes generate a primitive sublattice of $\mathrm{Pic}(\bar{X}_m)$. Indeed the map given by intersection with the three boundary components is onto $\mathbb{Z}^3$, as is already seen on $E_1, E_2, E_3$. Unimodularity of the intersection form therefore identifies the quotient by the boundary classes with $\Lambda^\vee$. The units of $\bar{U}_m$ are constant, and the boundary localization sequence identifies this quotient with $\mathrm{Pic}(\bar{U}_m)$. □

The period formulas are equivariant. More explicitly, if $\epsilon_i = \epsilon_i(g)$, then

$$g(\eta) = \epsilon_1\epsilon_2\epsilon_3\epsilon_4\eta^{\epsilon_4}, \qquad g(h) = \epsilon_4 h + \sum_{i=1}^{3} \frac{\epsilon_i - \epsilon_4}{2} e_i. \tag{5.4}$$

These identities also make the ensuing finite computation independent of any unspecified geometric Galois action.

5.2. **A finite module computing the complete group.** For a positive integer $n$ define the finite Galois module

$$T_n = \frac{\{(\lambda, t) \in \Lambda \times \overline{k}^* : t^n = \phi(\lambda)\}}{\{(n\lambda, \phi(\lambda)) : \lambda \in \Lambda\}}. \tag{5.5}$$

The group law is addition in the first coordinate and multiplication in the second. It fits into an exact sequence

$$0 \longrightarrow \mu_n \longrightarrow T_n \longrightarrow \Lambda/n\Lambda \longrightarrow 0. \tag{5.6}$$

Every element is killed by $n$, and $T_n$ is free of rank five over $\mathbb{Z}/n$.

**Theorem 5.2.** *The inclusion* $\mathrm{Br}(k) \to \mathrm{Br}(U_m)$ *is injective and admits a noncanonical retraction. For every* $n \geq 1$*, there is an isomorphism*

$$(\mathrm{Br}(U_m)/\mathrm{Br}(k))\,[n] \simeq \frac{\mathrm{Hom}_{G_k}(T_n, \mathbb{Z}/n)}{(\Lambda^\vee)^{G_k}/n(\Lambda^\vee)^{G_k}}. \tag{5.7}$$

*The denominator maps to the numerator by $\ell(\lambda,t)=\ell(\lambda) \bmod n$. Taking the direct limit, with the maps induced by the Kummer sequences, computes the complete group modulo constants.*

More explicitly, if $n \mid N$ and $d=N/n$, the reduction map $T_N \to T_n$ sends $(\lambda,t)$ to $(\lambda,t^d)$. The transition on the numerator sends a functional $F$ to $dF \circ \mathrm{red}$ with values in $\mathbf{Z}/N$; on the denominator it sends $\ell \bmod n$ to $d\ell \bmod N$. These maps specify the direct limit, so evaluation at a rational point gives

$$\mathrm{Br}(U_m) \simeq \mathrm{Br}(k) \oplus \varinjlim_{n} \frac{\mathrm{Hom}_{G_k}(T_n, \mathbf{Z}/n)}{(\Lambda^\vee)^{G_k}/n(\Lambda^\vee)^{G_k}}.$$

*Proof.* Evaluation at the affine point in Lemma 2.1 splits constants. Lemma 2.2 gives

$$\mathrm{Pic}(U_m) = \mathrm{Pic}(\bar{U}_m)^{G_k} = (\Lambda^\vee)^{G_k}. \tag{5.8}$$

Over $\overline{k}$, the relative cohomology sequence of $(\bar{X}_m, \bar{\mathscr{D}})$ gives

$$0 \longrightarrow H^1(\bar{\mathscr{D}}, \mu_n) \longrightarrow H^2_c(\bar{U}_m, \mu_n) \longrightarrow \ker\Big[\mathrm{Pic}(\bar{X}_m)/n \longrightarrow (\mathbb{Z}/n)^3\Big] \longrightarrow 0.$$

Here $H^1(\bar{X}_m, \mu_n)=0$, $\mathrm{Br}(\bar{X}_m)=0$, $H^1(\bar{\mathscr{D}}, \mu_n)=\mu_n$, and the last kernel is $\Lambda/n\Lambda$: the integral degree map is onto, as checked in Proposition 5.1. To identify the extension, apply Kummer to the relative cohomology complex of $(\bar{X}_m, \bar{\mathscr{D}})$. For $\mathbf{G}_{\mathrm{m}}$, the unit groups of $\bar{X}_m$ and $\bar{\mathscr{D}}$ agree, both geometric Brauer groups vanish, and the degree map from the Picard group is onto. The relevant relative Picard complex therefore reduces to $[\Lambda \xrightarrow{\phi} \overline{k}^*]$ in degrees one and two: the quotient complex $[\mathbf{Z}^3 \xrightarrow{1} \mathbf{Z}^3]$ is acyclic. Relative Kummer on this complex gives exactly (5.6). Concretely, a class $\lambda \in \Lambda$ restricts to the degree-zero boundary line bundle $\phi(\lambda)$; lifting modulo $n$ amounts to choosing an $n$-th root $t$ of this boundary class. Changing the lift by $n\nu$ changes $t$ by $\phi(\nu)$, precisely the relation in (5.5). Thus, equivariantly,

$$H^2_c(\bar{U}_m, \mu_n) \simeq T_n. \tag{5.9}$$

Poincaré duality [3], with pairing valued in $H^4_c(\bar{U}_m, \mu_n^{\otimes 2}) = \mathbb{Z}/n$, yields

$$H^2(\bar{U}_m, \mu_n) \simeq \mathrm{Hom}(T_n, \mathbb{Z}/n). \tag{5.10}$$

Dualizing (5.6) exhibits the geometric Kummer sequence

$$0 \longrightarrow \Lambda^\vee/n \longrightarrow \mathrm{Hom}(T_n, \mathbb{Z}/n) \longrightarrow \mathbb{Z}/n(-1) \longrightarrow 0.$$

The group $\mathrm{Pic}(\bar{U}_m)$ is torsion-free and the geometric units are constant; hence $H^1(\bar{U}_m, \mu_n)=0$. In the Hochschild–Serre spectral sequence for $\mu_n$, the only possible obstruction to lifting $H^2(\bar{U}_m, \mu_n)^{G_k}$ is its differential into $H^3(k, \mu_n)$. The injectivity established above makes this differential zero. We obtain

$$0 \longrightarrow H^2(k, \mu_n) \longrightarrow H^2(U_m, \mu_n) \longrightarrow H^2(\bar{U}_m, \mu_n)^{G_k} \longrightarrow 0.$$

Quotienting by first Chern classes and using (5.8) gives (5.7) for $\mathrm{Br}(U_m)[n]/\mathrm{Br}(k)[n]$. The retraction of constants identifies this group with the $n$-torsion of the quotient, proving the theorem. □

5.3. **Five-dimensional congruences and the transcendental part.** Formula (5.7) can be evaluated using ordinary finite Galois theory and linear congruences. The elements $e_1, e_2, e_3, h$ form a basis of $\Lambda$. Choose $q^n = -1$, $u^n = \eta$, and a primitive $n$-th root $\zeta$. The field

$$K_n = L(\mu_{2n}, u)$$

is Galois over $k$, by (5.4). The module $T_n$ has basis

$$\tau = (0, \zeta), \quad v_i = (e_i, q)\ (1 \le i \le 3), \quad v_h = (h, u).$$

For $g \in \mathrm{Gal}(K_n/k)$ put $a_i(g) = (\epsilon_i(g) - \epsilon_4(g))/2$ and define residues $c_i(g), c_h(g) \in \mathbb{Z}/n$ by

$$\zeta^{c_i(g)} = \frac{g(q)}{q^{\epsilon_i(g)}}, \qquad \zeta^{c_h(g)} = \frac{g(u)}{q^{\sum_{i=1}^{3} a_i(g)} u^{\epsilon_4(g)}}.$$

The ratios are $n$-th roots of unity by equivariance of $\phi$. Writing $g(\zeta) = \zeta^{\chi_n(g)}$, the action is

$$\begin{aligned} g(\tau) &= \chi_n(g)\tau, \\ g(v_i) &= \epsilon_i(g) v_i + c_i(g)\tau, \\ g(v_h) &= \epsilon_4(g) v_h + \sum_{i=1}^{3} a_i(g) v_i + c_h(g)\tau. \end{aligned}$$

Thus an invariant functional is a vector $(t, x_1, x_2, x_3, x_h) \in (\mathbb{Z}/n)^5$ satisfying, for all $g$,

$$\begin{aligned} (\chi_n(g) - 1)t &= 0, \\ (\epsilon_i(g) - 1)x_i + c_i(g)t &= 0 \quad (1 \le i \le 3), \\ (\epsilon_4(g) - 1)x_h + \sum_{i=1}^{3} a_i(g) x_i + c_h(g)t &= 0. \end{aligned}$$

Taking these vectors modulo the denominator in (5.7) is an explicit computation of the whole quotient, including extension data between its algebraic and transcendental parts.

The geometric Brauer group is $\mathrm{Br}(\bar{U}_m) \simeq \mathbb{Q}/\mathbb{Z}(-1)$, in agreement with [2, Proposition 2.4]. In the finite calculation, the restriction of a functional to $\mu_n \subset T_n$ is exactly its geometric Brauer image. The images of these finite calculations, as the modulus varies, compute the transcendental quotient. One must distinguish the order of a geometric image from the order of its lift: an image of order $n$ need not have a lift of order $n$ modulo constants. The algebraic quotient is killed by 4. Indeed $\Lambda^\vee$ is the even-sum sublattice of the diagonal sign module $\mathbb{Z}^4$; its quotient in that module is $\mathbb{Z}/2$, and $H^1(k, \mathbb{Z}^4)$ is killed by 2. The cohomology sequence therefore shows that $H^1(k, \Lambda^\vee)$ is killed by 4. It follows that the transcendental quotient contains an element of order $n$ if and only if the congruences with modulus $4n$ have a solution whose value on $\tau$ is $t = 4$. To see this, a geometric image of order $n$ has a lift killed by $4n$; its value on $\mu_{4n}$ is $4u$ with $u$ prime to $n$, and multiplying by an inverse of $u$ modulo $n$ changes that value to 4. The converse is immediate. This criterion implies $\mu_n \subset k$. If the algebraic quotient has exponent $e$, one may instead use modulus $en$ and value $t = e$.

**Corollary 5.3.** *If $n$ is odd, the transcendental quotient contains a subgroup of order $n$ if and only if*

$$\mu_n \subset k \qquad \text{and} \qquad \eta \in L^{*n}.$$

*For arbitrary $n$, necessary conditions are $\mu_n \subset k$ and $-1, \eta \in L^{*n}$. In all cases the five-dimensional congruences with modulus $4n$ and value $t = 4$ give a necessary and sufficient criterion.*

*Proof.* Over $L$, the lattice action is trivial and the algebraic quotient vanishes. Consequently a geometric image of order $n$ has a lift of order $n$ modulo constants there. An invariant functional on $T_n$ restricting to 1 on $\mu_n$ exists precisely when the Kummer characters of its four period values vanish. By (5.2), this means $-1, \eta \in L^{*n}$. Restriction proves necessity. Conversely, under these conditions there is a Brauer class over $L$ with geometric image of order $n$. When $n$ is odd, corestriction multiplies its geometric image by the power of two $[L : k]$, which is invertible modulo $n$; a suitable multiple therefore has geometric image of order $n$ over $k$. Finally, $-1$ is automatically an $n$-th power for odd $n$. □

**Remark 5.4.** When $a = b = c = 1$, the calculation specializes to the ordinary Markoff surface. In this case [1, Theorem 4.7] simplifies the transcendental criterion for every $n$ to $\mu_n \subset k$ and $-1, \eta \in k(\sqrt{m}, \sqrt{m-4})^{*n}$. For general coefficient twists the finite Galois congruences retain the additional descent conditions; the ordinary Markoff criterion should not be transferred to the twisted surface without checking them.

## 6. Consequences and examples

### 6.1. A restriction on the two-primary geometric image.

**Proposition 6.1.** *If the image of $\mathrm{Br}(U_m)$ in $\mathrm{Br}(\overline{U}_m)$ contains an element of order $2^s$, where $s \geq 1$, then*

$$\mu_{2^{s+1}} \subset k(\sqrt{a\Delta}), \qquad \mu_{2^{s+1}} \subset k(\sqrt{b\Delta}), \qquad \mu_{2^{s+1}} \subset k(\sqrt{c\Delta}). \tag{6.1}$$

*In particular, if $\sqrt{-1} \notin k$, a nonzero two-primary geometric image is possible only if $-a\Delta, -b\Delta, -c\Delta$ are squares in $k$.*

*Proof.* Let a class have geometric image of order $2^s$. After removing constants using Lemma 2.1, choose an integer $N$ killing the class. Theorem 5.2 represents it by an invariant functional $F : T_N \to \mathbf{Z}/N$ whose value on $\tau$ has order $2^s$. Choose $q = \zeta_{2N}$ and $\zeta_N = \zeta_{2N}^2$ in the matrix calculation. If $g$ fixes the character $\chi_i$, for $1 \leq i \leq 3$, then

$$g(v_i) = v_i + \frac{\chi_{2N}^{\mathrm{cyc}}(g) - 1}{2}\tau.$$

Here $\chi_{2N}^{\mathrm{cyc}}$ is the cyclotomic character modulo $2N$, and its values are odd. Invariance of $F$ implies that the displayed coefficient is divisible by $2^s$, since $F(\tau)$ has that exact order. Thus $g$ acts trivially on $\mu_{2^{s+1}}$. Applying this to each kernel of $\chi_i$ proves (6.1).

If $\sqrt{-1} \notin k$, each of the three quadratic fields in (6.1) must equal $k(\sqrt{-1})$. Equality of quadratic extensions gives $[a\Delta] = [b\Delta] = [c\Delta] = [-1]$, as asserted. □

**Corollary 6.2.** *Let $k = \mathbf{Q}$. If $a, b, c$ do not all have the same square class, then*

$$\mathrm{Br}(U_m) = \mathrm{Br}_1(U_m).$$

*Consequently the complete quotient* $\mathrm{Br}(U_m)/\mathrm{Br}(\mathbf{Q})$ *is given by* (1.6) *in this case.*

*Proof.* The invariant subgroup of $\mathbf{Q}/\mathbf{Z}(-1)$ over $\mathbf{Q}$ has order two: an invariant element of order $n$ requires $\mu_n \subset \mathbf{Q}$. Any nonzero geometric image therefore has order two. Proposition 6.1 would imply $[a\Delta] = [b\Delta] = [c\Delta] = [-1]$, hence $[a] = [b] = [c]$, contrary to the hypothesis. □

6.2. **The ordinary Markoff specialization.** Suppose $ab$ and $ac$ are squares in $k$. Choose $r_1^2 = bc$, $r_2^2 = ac$ and $r_3 = r_1 r_2/c$, so that $r_3^2 = ab$ and $r_1 r_2 r_3 = abc$. The substitution

$$X = x/r_1, \qquad Y = y/r_2, \qquad Z = z/r_3$$

is defined over $k$ and identifies $U_m$ with the ordinary Markoff surface of parameter $M = m/(abc)$. Put $D = M - 4$ and $K = k(\sqrt{M}, \sqrt{D})$, and choose $\eta = (\sqrt{M} - \sqrt{D})/2$ as before. The algebraic quotient is

$$\mathrm{Br}_1(U_m)/\mathrm{Br}(k) \simeq \begin{cases} 0, & D \in k^{\times 2}, \\ (\mathbf{Z}/2)^2, & D \notin k^{\times 2},\ MD \in k^{\times 2}, \\ (\mathbf{Z}/2)^4, & D \notin k^{\times 2},\ M \in k^{\times 2}, \\ (\mathbf{Z}/2)^3, & [M], [D] \text{ are independent.} \end{cases} \tag{6.2}$$

These are the formulas of [1, Theorem 3.4].

For the transcendental quotient, define

$$\mathcal{I} = \{n \geq 1 : \mu_n \subset k \text{ and } -1, \eta \in K^{\times n}\}. \tag{6.3}$$

Theorem 4.7 of [1] gives

$$\mathrm{Br}(U_m)/\mathrm{Br}_1(U_m) \simeq \varinjlim_{n \in \mathcal{I}} \mathbf{Z}/n. \tag{6.4}$$

The transition maps are the natural embeddings in $\mathbf{Q}/\mathbf{Z}$. The sign difference between the root used there and our $\eta$ does not change (6.3), because $-1$ is also required to be an $n$-th power. Over a number field the group in (6.4) is finite cyclic. Formula (5.7) also determines the extension of this cyclic group by (6.2).

**Remark 6.3.** The square-class classification of the algebraic group applies to every coefficient twist. The root criterion (6.3) applies here because the displayed change of variables is defined over $k$. In general the Galois congruences in Section 5 must be used for descent; adjoining the square roots of the coefficients alone does not compute the geometric image over the original field.

6.3. **Examples.**

**Example 6.4.** Let $k = \mathbf{Q}$ and $(a, b, c) = (1, 2, 3)$. For $m = 30$ the discriminant is $\Delta = 6$, and

$$(\chi_1, \chi_2, \chi_3, \chi_4) = ([6], [3], [2], [5]).$$

All four characters are nonzero, their span has dimension three, and $\chi_1 + \chi_2 + \chi_3 = 0$ is an odd-weight relation. Hence

$$\mathrm{Br}(X_{30})/\mathrm{Br}(\mathbf{Q}) = \mathbf{Z}/2, \qquad \mathrm{Br}(U_{30})/\mathrm{Br}(\mathbf{Q}) = \mathbf{Z}/4.$$

The second equality uses Corollary 6.2. For $m = 48$, the character tuple is $([6], [3], [2], [2])$, and the same argument gives

$$\mathrm{Br}(X_{48})/\mathrm{Br}(\mathbf{Q}) = \mathbf{Z}/2, \qquad \mathrm{Br}(U_{48})/\mathrm{Br}(\mathbf{Q}) = \mathbf{Z}/4 \oplus \mathbf{Z}/2.$$

In both cases the nonzero projective class is the double of an affine class of order four modulo constants.

**Example 6.5.** Let $k = \mathbf{R}$, $a = b = 1$, $c = -1$ and $m = -5$. Then $\Delta = -1$, and the four characters are $([-1], [-1], 0, 0)$. Thus $N = 2$, $d = 1$, and

$$\mathrm{Br}_1(U_m)/\mathrm{Br}(\mathbf{R}) \simeq \mathbf{Z}/2.$$

In fact this is the complete quotient. Here $M = 5$, $D = 1$, and $\eta = (\sqrt{5} - 1)/2$ can be chosen positive. For every $N$, complex conjugation fixes $e_3$ and sends $q = \exp(\pi i/N)$ to $q^{-1}$. Consequently $g(v_3) = v_3 - \tau$. Every invariant functional on $T_N$ kills $\tau$, so its geometric Brauer image is zero. Notice that all period values have arbitrary roots in the splitting field $L = \mathbf{C}$; the failure here is descent to $\mathbf{R}$.

**Example 6.6.** If the four square classes in (1.5) are independent, then the projective quotient and the algebraic affine quotient are both $\mathbf{Z}/2$, generated by $(x^2 - 4bc, b\Delta)$. Over $\mathbf{Q}$ the coefficient classes cannot all agree in this situation. Corollary 6.2 therefore shows that the complete affine quotient is also $\mathbf{Z}/2$.

## References


[1] J.-L. Colliot-Thélène, D. Wei and F. Xu, *Brauer–Manin obstruction for Markoff surfaces*, Ann. Sc. Norm. Super. Pisa Cl. Sci. (5) **21** (2020), 1257–1313.

[2] D. Loughran and V. Mitankin, *Integral Hasse principle and strong approximation for Markoff surfaces*, Int. Math. Res. Not. IMRN (2021), no. 18, 14086–14122.

[3] J. S. Milne, *Étale cohomology*, Princeton Mathematical Series, vol. 33, Princeton University Press, Princeton, NJ, 1980.

[4] B. Poonen, *Rational points on varieties*, Graduate Studies in Mathematics, vol. 186, American Mathematical Society, Providence, RI, 2017.



Academy of Mathematics and System Science, CAS, Beijing 100190, P. R. China *and* School of Mathematical Sciences, University of CAS, Beijing 100049, P. R. China

*Email address*: dshwei@amss.ac.cn